

# Year 8 Science – Booklet 17: Physics

## Motion & Pressure — ANSWERS

### Graphs of Motion – Quick Test (p.89)

#### 1. On a distance-time graph, what do these show?

- a). A horizontal line: the object is not moving (stationary).
- b). A steeply sloping straight line: the object is moving at a high, constant speed.
- c). A straight line sloping just a little: the object is moving at a low, constant speed.

#### 2. On a speed-time graph, what do these show?

- a). A horizontal line: the object is moving at a constant speed (or is stationary, if the line is along the time axis).
- b). A straight line sloping steeply upwards: the object is accelerating quickly (a large acceleration).
- c). A straight line sloping gently downwards: the object is decelerating slowly (a small deceleration).

### Graphs of Motion – Section A (5 marks)

- 1. d) moving at a constant speed
- 2. c) its speed
- 3. b) accelerating
- 4. a) moving at a high speed
- 5. a) moving with a high acceleration

### Graphs of Motion – Section B (6 marks)

#### 1. Walker's distance-time graph (A→B→C→D)

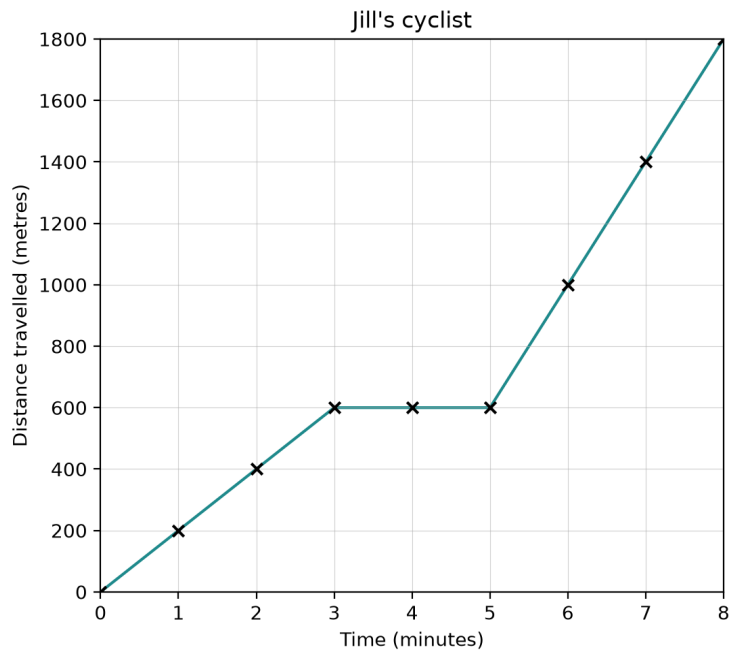
- a). B to C (the flat section).
- b).  $250 - 125 = 125$  seconds
- c). A to B (the steepest section – 500 m in 125 s, faster than C to D's 500 m in 250 s).
- d). speed =  $500 \text{ m} \div 125 \text{ s} = 4 \text{ m/s}$
- e). **1000 m** (the total distance reached by point D).
- f). average speed =  $1000 \text{ m} \div 500 \text{ s} = 2 \text{ m/s}$

### Graphs of Motion – Section C (20 marks)

#### 1. Car's speed-time graph (A–F: 60 km/h, then stop, then 90 km/h)

- a). C to D (the section at zero speed).
- b).  $1.0 - 0.5 = 0.5$  hours
- c). E to F (at 90 km/h, faster than the 60 km/h A–B section).
- d). **90 km/h**
- e). distance =  $(60 \text{ km/h} \times 0.5 \text{ h}) + (0 \text{ km}) + (90 \text{ km/h} \times 1.0 \text{ h}) = 30 + 90 = 120 \text{ km}$
- f). average speed =  $120 \text{ km} \div 2.0 \text{ h} = 60 \text{ km/h}$

#### 2. Jill's cyclist (distance-time table, 0–8 min)



- b). From minute 3 to minute 5, the distance stays at 600 m, so the cyclist was stationary for **2 minutes**.
- c). The distance increases by 400 m in each of the intervals 5–6, 6–7 and 7–8 minutes – these three are tied for fastest.
- d). Greatest speed: 400 m in 1 minute (60 s) =  $400 \div 60 \approx \mathbf{6.67 \text{ m/s}}$
- e). average speed =  $1800 \text{ m} \div (8 \times 60 \text{ s}) = 1800/480 = \mathbf{3.75 \text{ m/s}}$

## Pressure – Quick Test (p.97)

1. By applying a force over a small area (concentrating the force).
2. By spreading the force over a larger area.
3. The weight of the shopping is concentrated onto the small area of the thin handles, creating a high pressure on your hands.
4. Pascals (Pa).
5. A sharp knife has a much smaller (thinner) blade edge area, so the same force creates a much higher pressure, making it easier to cut through.
6.  $P = F/A = 50 \div 2.5 = \mathbf{20 \text{ Pa}}$
7. Area =  $4 \times 2 = 8 \text{ m}^2$ .  $P = 4000 \div 8 = \mathbf{500 \text{ Pa}}$
8.  $F = P \times A = 40 \times 2 = \mathbf{80 \text{ N}}$
9. At the bottom of the lake (the greatest depth) – pressure increases with depth.

## Pressure – Section A (5 marks)

1. c) someone wearing snowshoes
2. c) 16 Pa
3. b) 50 N
4. b)  $5 \text{ m}^2$
5. b) increases with depth

## Pressure – Section B (13 marks)

### 1. True or false (7 marks)

- a) Thin handles create high pressure, making it easier to carry – **False** (high pressure makes it more painful, not easier)
- b) Snowshoes spread force to create low pressure, stopping you sinking – **True**
- c) Pressure on a nail is greatest on the head – **False** (it's greatest at the sharp point, the smallest area)
- d) Sharp knives create higher pressures than blunt knives – **True**
- e) Press-ups on palms create the same pressure as on fingertips – **False** (fingertips have a much smaller area, so create much higher pressure for the same weight)
- f) Camels have large feet to grip the sand better – **False** (large feet spread their weight to reduce pressure, stopping them sinking – not about grip)
- g) More likely to fall through a frozen lake standing up than lying down – **True** (standing concentrates your weight on your feet; lying down spreads it over a larger area)

### 2. Calculate the pressure

- a). 50 N over 25 m<sup>2</sup> =  $50/25 = 2$  Pa
- b). 80 N over 16 m<sup>2</sup> =  $80/16 = 5$  Pa
- c). 25 N over 2.5 m<sup>2</sup> =  $25/2.5 = 10$  Pa

### 3. Calculate the force

- a). over 10 m<sup>2</sup> for 3 Pa:  $F = 3 \times 10 = 30$  N
- b). over 4 m<sup>2</sup> for 5 Pa:  $F = 5 \times 4 = 20$  N
- c). over 16 m<sup>2</sup> for 1.5 Pa:  $F = 1.5 \times 16 = 24$  N

## Pressure – Section C (12 marks)

### 1. 600 N crate, faces A, B, C (3 m × 2 m × 1 m)

The exact face each letter refers to is hard to confirm precisely from this scan – this reading assumes A = the 2 m × 1 m side face (2 m<sup>2</sup>), B = the 3 m × 1 m side face (3 m<sup>2</sup>), and C = the 3 m × 2 m base (6 m<sup>2</sup>); check against your own diagram.

- a). Face A (2 m<sup>2</sup>):  $P = 600/2 = 300$  Pa
- b). Face B (3 m<sup>2</sup>):  $P = 600/3 = 200$  Pa
- c). Face C (6 m<sup>2</sup>):  $P = 600/6 = 100$  Pa
- d). Face C should be in contact with the lawn – it has the largest area, so it creates the lowest pressure (least likely to damage the grass).

### 2. Drawing pin diagram

- a). Mark H at the sharp point of the pin (smallest area, force concentrated = high pressure).
- b). Mark L at the flat, broad head of the pin (largest area, force spread out = low pressure).